

# Pessimistic rumination in popular songs and newsmagazines predict economic recession via decreased consumer optimism and spending \*

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Lyrics of the top 40 U.S. songs of each year from 1955 to 1989 were content-analysed for two depressive psychological traits: rumination about bad events, and pessimistic explanatory style. The cover story captions of *Time* magazine for those years were also analysed for rumination. Increased pessimistic ruminations in popular music predicted changes in the American media and public's view of real world events with a one- to two-year lead time. It predicted increased rumination about bad events in *Time* magazine, and increased pessimism about the economy in nationwide consumer surveys. Using two-stage least squares to solve a system of simultaneous equations, pessimistic rumination in songs and rumination in *Time* predicted changes in consumer optimism, which in turn predicted personal consumption expenditures and GNP growth. Although pessimistic rumination in songs was an indirect predictor of GNP growth, it may provide early warning of recessions, since its two-year moving average correlated highly ( $-0.63$ ) with the moving average of GNP change in the subsequent two years. The implication of recent high levels of pessimistic rumination in songs (in 1989) for a recession are discussed.

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‘A mathematician’s mood exercises no influence upon his solution of an algebraic equation, but it does affect his opinion about the advisability of buying the bonds offered him.’

Wesley Clair Mitchell (1959: 5)

‘As to receiving a new kind of music one should be specially cautious, as endangering the whole; for never...are the measures of music altered without affecting the most important laws of state...It does nothing...but by gradually insinuating itself into it [public], insensibly flow into their manners and pursuits.’

Plato (389-369 B.C./1900: 108)

## Introduction

Consumer optimism, assessed in nationwide surveys, is a robust leading indicator of the onset and abatement of economic recessions. Some economists have explained mass shifts in optimism as a reaction to the economy (Shapiro 1972; Vanden Abeele 1983). I propose that in part, these shifts reflect and are influenced by culturally transmitted fantasies of hope and despair. Is our national mood entrained by the rhythm of popular culture? If so, hope and despair in the culture should predict changes in consumer optimism and economic growth.

Since Katona and others in the Economic Behavior Program at the University of Michigan began studying consumer optimism in the 1950s, it has become increasingly evident that national mood changes before a recession begins – typically nine months in advance (Katona 1960, 1980). Surveys conducted two to four times a year until 1977, and monthly since 1978, have asked a nationwide sample of Americans about their current and expected financial situation, and that of the economy. Responses are combined into an Index of Consumer Sentiment (ICS), which the U.S. Commerce Department has used since 1978 as a leading indicator of economic activity. The ICS forewarned us of the last seven recessions, and sounded one false alarm before the minirecession of 1966–67. The ICS predicts changes in durable goods purchases, such as automobiles, by American consumers. These expenditures reflect the use of discretionary income, and provide an advance indicator of turning points in the economy.

These findings have been replicated by continuing surveys in other countries (Katona and Strümpel 1978; Praet 1985; Williams and Defris

1981). A nation's generalized (non-economic) hopes and fears also predict changes in GNP (Noelle-Neumann 1984, 1989).

Does consumer sentiment predict the economy above and beyond traditional macroeconomic variables? Some have found that the ICS can be explained largely by traditional macroeconomic variables: that consumer sentiment reacts to the economy, but does not contribute over and above to predicting economic change (Shapiro 1972; Vanden Abeele 1983). In contrast, Katona and others (Adams and Klein 1972; Pickering 1977; Praet 1985; Van Raaij and Gianotten 1990; Williams and Defris 1981) have shown that changes in consumer sentiment are at least in part independent of economic change and contribute to predicting macroeconomic trends.

What drives these changes in consumer sentiment? I propose that they are attributable in part to cognitive changes in consumers' evaluations of bad events. Two cognitive dispositions, which are also risk factors for psychological depression, are proposed: (1) rumination about bad events, and (2) explanatory style for bad events. If these dispositions, which affect the formation of expectancies at an individual level, are transmitted via the mass media, popular culture, and other channels of social interaction, then they also might relate to the formation of expectancies on a mass scale (Zullow et al. 1988).

One disposition is a pessimistic explanatory style (Abramson et al. 1978). Pessimists explain bad events as due to causes that are *stable* ('it's going to last forever'), *global* ('it's going to affect everything we do'), and *internal* ('it's our fault'). Optimists, on the other hand, explain bad events as due to causes that are *unstable* ('it's temporary'), *specific* ('it only affects this one problem') and *external* to themselves ('it's the situation we're in'). Pessimistic causal explanations lead us to anticipate that the cause of a bad event will persist and pervade other areas – to extrapolate from one event so as to expect more. Optimists see bad events as caused by factors that would not be expected to lead to continued bad events. An optimistic explanatory style predicts freedom from depression, persistence in the face of obstacles, and high achievement (Peterson and Seligman 1984; Robins 1988).

An example of a pessimistic explanation was in the 1963 song 'Puff the Magic Dragon', scored for pessimism from the perspective of the dragon who is the protagonist. In this song Jackie, a human playmate of Puff's, has grown up and no longer comes to play. Puff's bad event is: 'His head was bent in sorrow, green scales fell like rain. Puff no

longer went to play along the cherry lane... Puff that mighty dragon sadly slipped into his cave'. The cause of this event is 'Without his lifelong friend Puff could not be brave'. This is a stable and global cause, likely to persist and affect many areas of Puff's life, and is also moderately internal to Puff as he is unable to be brave without his friend.

In contrast, optimistic explanations are found in the 1957 Everly Brothers' song 'Wake Up Little Susie'. After a teenage couple stays out past curfew, the boy worries 'Our goose is cooked, our reputation is shot'. His causal explanation for this is 'You fell asleep', i.e., Susie fell asleep at the movie. In turn, Susie fell asleep because 'The movie wasn't so hot, it didn't have much of a plot'. Both of these explanations are less stable, global, and internal than that for Puff the Magic Dragon's bad event. One needn't expect that Susie and her boyfriend will continue to get into trouble, because future movies may be more exciting and hence Susie will remain awake (unstable and specific cause). However, Puff will probably continue to be shattered because 'a dragon lives forever but not so little boys', and 'without his lifelong friend Puff could not be brave' (stable and global).

The other cognitive disposition, rumination (Kuhl 1981, 1984; Zullow 1984), is how frequently we dwell on problems and evaluate situations negatively. Rumination indexes how frequently we engage in causal and other non-action-oriented analyses of bad events. Ruminators take longer to recuperate from bad events because their perserverative thinking interferes with problem-solving (Kuhl 1981, 1984). An example of a rumination was in the 1969 Creedence Clearwater song, 'Bad Moon Rising': 'I see the bad moon a-rising, I see trouble on the way. I see earthquakes and lightnin'. I see bad times today'.

The effect of rumination may be compounded by a pessimistic explanatory style, and vice versa. A pessimistic explanatory style, combined with rumination, disposes ordinary individuals to depression (Kammer 1983, 1984; Rholes 1989; Zullow 1984). A pessimistic explanatory style extrapolates bad events stably into the future and globally across situations. Ruminating also can generate negative expectations, via the mechanism of an 'availability heuristic' (Carroll 1978; Sherman et al. 1985). Imagining a potential bad event increases the subjective likelihood that it will occur, because a mental representation of the event is more readily available.

Not only does pessimistic rumination predict lowered mood, expect-

tations, and activity, but ruminating aloud to other people affects their mood (Pennebaker 1990). Therefore, pessimistic rumination might be propagated throughout society via popular culture, mass media, and other channels of social interaction. In this view, popular culture and mass media may reflect psychological trends, but also propagate and homogenize trends throughout the society. For example, popular songs that dwell on bad events may reflect a trend toward rumination among the public. Songs and other popular culture may then have a contagion effect, propagating rumination among individuals not previously inclined. This proposed model differs from one in which popular culture influences mass psychology, without reflecting changes in mass psychology.

Because causal explanations and ruminations are mental filters that color how we view real-world events, to get an early window into changes in the public's real-world outlook and behavior we can examine popular culture such as songs and movies that reflect our communal fantasy life rather than real life. Popular culture can forewarn us of changes in public sentiment not yet evident in public opinion polls. As the cognitive changes seen in popular culture influence our perceptions of real-world events, and the attention we pay to real-world bad events, changes in public opinion should occur. These attitudinal changes should in turn map into changes in our national political and economic life.

Pessimistic rumination may fluctuate due to:

(1) Age cohorts reared under different formative experiences. As a cohort nears its ages of critical influence on the socioeconomic development of a nation, its mentality will be increasingly felt in the popular culture and media.

(2) Homeostatic feedback mechanisms, such as those in game theoretic simulations of how non-equilibrium conditions in a society are corrected. Homeostatic mechanisms in a healthy society should correct affective deviations from the mean. If consumer sentiment were responsive to economic change only in a positive feedback loop, how would the nation's mood keep in balance? A society might then stray in a manic, financially speculative and debt-incurring direction or in a depressive, economically stagnant direction.

(3) Current events, both political and economic.

(4) Social learning of ruminative styles and of defense mechanisms that encourage or discourage rumination. For example, the spread of *denial* as a defense mechanism would discourage rumination. Recurrent boomlets of popularity of songs such as 'Pack up Your Troubles in Your Old Kit Bag and Smile, Smile, Smile', 'Bye Bye Blackbird', and 'Don't Worry, Be Happy' show that Americans are susceptible to such periods of denial.

Joseph Schumpeter (1964) and Wesley Clair Mitchell (1959) were concerned with the economic effects of fluctuations in optimism, and proposed how changes in optimism might spread through a society. They stress that optimistic and pessimistic styles may lead to characteristic behavioral biases, which may promote a swing back in the other direction. They envision a negative feedback loop in society's optimism and business activity, which generates a business cycle. For example, easy money is an optimistic policy, but can have side effects. At some point the purveyors of an optimistic policy may find reality constraining them toward greater pessimism and tighter credit. Donald Trump's building of the Taj Mahal casino at a time of declining business in Atlantic City – driving him deeper into debt – is an example of an error of optimism. Similarly, an optimistic assessment of consumer demand by producers might lead to an inventory buildup, which would then promote increased pessimism and a cutback in production and employment.

Let me specify the model I want to test. The initial stage in this model is an increase in society's pessimistic rumination, for whatever causes. This increase will first be evident, I hypothesize, in the feared fantasies depicted in cultural products such as popular songs, movies, and novels. Then, the second step is that the disposition toward pessimistic rumination, depicted in fantasy material, should constitute a diathesis for perceiving real-world events through a negative cognitive filter. The reformulated diathesis-stress model of learned helplessness (Abramson et al. 1978) specifies such a process by which a pessimistic explanatory style, as applied to explain actual bad events, leads to negative expectations. Bad economic events such as the 1973 OPEC oil embargo or the Iraqi invasion of Kuwait, when viewed by the public through the prism of an increased disposition toward pessimistic rumination, should prompt consumers to develop increasingly negative expectations and mood. When consumers become more pessimistic

about the economy and their personal finances, personal consumption expenditures should fall. The economy should then tilt toward recession, as businesses also fall prey to negative expectations and adjust their inventories to slackening consumer activity.

## Method

### *Materials*

#### *Songs*

The top 40 U.S. songs of each year from 1955 to 1989 were selected, according to a compilation of *Billboard* magazine ratings (Whitburn 1990) based on nationwide airplay and sales. These songs reached the top of the weekly Hot 100 *Billboard* chart: usually a peak position of No. 1 or No. 2. The Hot 100 chart has been compiled consistently since 1955. Songs that were revived as a hit were counted for both years. For example, the gloomy song 'If You Don't Know Me By Now (You Will Never Know Me)' was number one in 1972 and again in 1989. Songs that were instrumental (e.g. 'Theme from A Summer Place') or foreign language (e.g. 'La Bamba') were excluded. This left 1,365 songs to be analysed. Of these, lyrics were obtained for 1,344 (98.5%). Lyrics were obtained from sheet music, periodicals such as *Song Hit Magazine* that publish lyrics, or were transcribed from the hit recording.

#### *Magazines*

The captions on 1,827 weekly covers of *Time* magazine from 1955 to 1989 were analysed to assess changes in rumination in the mass media. The cover caption of *Time* rarely offers a causal explanation, so only rumination was rated. *Time* was selected because during this period it was one of the most popular newsmagazines in the country. *Time*'s cover portrayal of business versus other leaders was used in a previous study (Cover and Johnson 1976) to assess correlates of changes in the business cycle.

#### *Index of Consumer Sentiment*

The published figures for the Index of Consumer Sentiment (Survey Research Center 1990) for 1955 to 1989 were used as a measure of consumer optimism. To obtain annual scores, the average ICS was

calculated across all surveys within a year. The Consumer Research Center of the Conference Board Inc., New York has also surveyed consumer confidence monthly since 1969, and its (unpublished) data were used to validate independently the relationship between pessimistic rumination (PessRum) and consumer sentiment.

#### *Personal consumption expenditures and GNP growth*

The annual percentage growth in personal consumption expenditures and GNP, adjusted for inflation, were taken from the *Economic Report of the President* (1989).

#### *Procedure*

##### *Raters*

One rater rated each song for rumination. A second rater rated 349 (26%) of the songs as a check on interrater reliability. All causal explanations for bad events were extracted from the songs. Two other raters rated these explanations for stability, globality, and internality of the cause.

One rater rated each *Time* cover for the presence or absence of rumination. A second rater independently rated the covers as a check on reliability.

Raters were originally trained with practice materials, and achieved high reliability with experts' ratings (over 90% agreement on the presence or absence of rumination in a sentence). Raters of causal explanations were blinded to the song's identity, by reading not the entire song but only the phrase containing the bad event and its explanation.

##### *Rumination ratings*

Rumination was content-analysed according to a system outlined in a manual (Zullow 1985) which follows the logic of the Kuhl (1981) questionnaire. For each sentence, raters decided if it contained a rumination – a negative description or evaluation of an event. For songs, the event had to be bad from the perspective of the singer (or the story's protagonist if the singer was a narrator) to rate as a rumination. For the *Time* covers, the event had to be bad from the perspective of the contemporaneous American public. If much of the public might regard the event as good, the cover did not rate for rumination. For

example, 'Democrats Lose in Senate Elections' does not rate for rumination because Republicans would see this as a good event. An event that is bad for an ally of the U.S. counts as a rumination, as do: natural disasters anywhere in the world, wars, and events that are bad for individuals with whom Americans identify (e.g., sports heroes and movie stars). A bad event's occurrence need not be accompanied by a ruminative description on the cover of *Time*. For example, Kennedy was assassinated at a time of high public confidence as measured in songs and consumer surveys. *Time*'s cover portrayed a leaderly Lyndon Johnson, with the non-ruminative caption 'President Johnson'. In contrast, 'Tragedy in Dallas' would have been a ruminative caption.

Rumination was rated for: (1) a focus on a negative event – the how, what, who, and where; (2) a causal explanation for a bad event – the why; and (3) negative emotion. A score for the rate of occurrence of ruminations (*Rum*) was derived by taking the percentage of sentences in a song that contained rumination. This score could range from 0 to 100%. For *Time* covers, a binary judgment was made regarding the presence or absence of rumination on the cover. Table 1 shows how the 1969 song 'Bad Moon Rising' was rated.

#### *Explanatory style ratings*

Songs were content-analysed according to the technique of Content Analysis of Verbatim Explanations (CAVE; Peterson et al., in press; Zullo et al. 1988). Causal explanations and the bad events explained were extracted from the text. To qualify for extraction, events had to be bad from the singer's perspective, or that of the story's protagonist (if the singer was narrating a story). Each explanation was counted only once, even if the lyric is sung more than once. Explanations were rated along 7-point scales for the dimensions of stable–unstable, global–specific, and internal–external. The stability, globality, and internality of the *causes* of an event was rated – not the event itself, nor its consequences. High ratings indicate high stability, globality, or internality, and low ratings indicate instability, specificity, or externality. The stable, global, and internal ratings were averaged across explanations for each song. These means were summed to obtain a composite pessimism (*Pess*) score with a possible range from 3 (most optimistic) to 21 (most pessimistic).

The cause was rated as highly stable (7) if likely to persist. It was rated as highly unstable (1) if it was past tense, or unlikely to continue.

Table 1

An example of rumination rated in a song: 'Bad Moon Rising' (1969).

| Sentence                                   | Rumination (present or absent?) |
|--------------------------------------------|---------------------------------|
| 1. I see a bad moon a-rising.              | Present                         |
| 2. I see trouble on the way.               | Present                         |
| 3. I see earthquakes and lightning.        | Present                         |
| 4. I see bad times today.                  | Present                         |
| (Chorus)                                   |                                 |
| 5. Don't go around tonight.                | Absent                          |
| 6. Well, it's bound to take your life.     | Present                         |
| 7. There's a bad moon on the rise.         | Present                         |
| (Verse)                                    |                                 |
| 8. I hear hurricanes a-blowing.            | Present                         |
| 9. I know the end is coming soon.          | Present                         |
| 10. I fear rivers overflowing.             | Present                         |
| 11. I hear the voice of rage and ruin.     | Present                         |
| (Repeat chorus)                            |                                 |
| (Verse)                                    |                                 |
| 15. Hope you got your things together.     | Absent                          |
| 16. Hope you are quite prepared to die.    | Present                         |
| 17. Looks like we're in for nasty weather. | Present                         |
| 18. One eye is taken for an eye.           | Present                         |
| (Repeat chorus twice)                      |                                 |

Note: 19/24 sentences contain rumination, hence rumination score = 79.2%.

The cause was rated as highly global (7) if it affected all aspects of the character's life: affiliative and achievement. It was rated as highly specific (1) if it affected only one narrow problem. The cause was rated as highly internal (7) if it blamed oneself, or one's ingroup. It was rated as highly external (1) if it blamed the situation or another person, and in between (4) if due equally to inner and outer forces.

### *Rater reliability*

Interrater reliability was satisfactory. For the 349 songs rated for rumination by two raters, the correlation between their overall scores for the songs was  $r = 0.90$  ( $p < 0.0001$ ). Cronbach's (1951) alpha for the two explanatory style raters was 0.65 for the composite pessimism score. Alphas were 0.83, 0.69, and 0.89 for the respective dimensions of stability, globality and internality.

For *Time* magazine, raters agreed 93.4% of the time that rumination was present or absent in the cover caption. Since there was a higher base rate probability of absence than presence of rumination in a

sentence, we can ask the frequency of agreement on the *presence* of rumination. This was calculated as the number of times *both* raters rated rumination as present, divided by the total number of covers rated for rumination by either rater. The level of agreement for presence of rumination was 77.9%.

### *Data reduction*

The percentage of sentences containing rumination was calculated for each song. If a chorus was repeated more than once, its sentences were counted as many times as they appeared. Fade-out lyrics were automatically counted twice. The rumination percentage score for same-year songs was averaged to yield a rumination percentage for the year (*RumSong*). This percentage varied from a low of 17.5% in the bicentennial year of 1976 to highs of 36.6% in the pre-recession years of 1973 and 1979, and 36.8% in 1971.

The stability, globality, and internality ratings for causal explanations from songs from the same year were averaged and summed to yield an annual pessimism score (*PessSong*) which could range from 3 (most optimistic) to 21 (most pessimistic). *PessSong* ranged from an optimistic low of 8.11 in 1963 – John F. Kennedy’s final year – to a pessimistic high of 12.59 in 1980 (surpassed during the very pessimistic second half of 1989).

To indicate the extent to which a year was high in rumination *and* pessimism, standardized *z* scores for the years 1955–1989 were calculated for rumination and pessimism. The *z* scores were summed, to create an index of pessimistic rumination (*PessRum*). Thus, a year that is one standard deviation above the mean for both rumination and pessimism receives a *PessRum* score of 2.0.

The percentage of ruminative *Time* covers in each year constituted the *RumTime* percentage. This ranged from a low of 2% in 1955–1957 to a high of 63% in 1989.

## **Results**

Results are divided into two sections: correlational results, and the two-stage least squares (2SLS) results for a structural equation model of the pessimistic rumination–GNP relationship.

Table 2

Pessimism and rumination in popular music and *Time* magazine, 1955–1989.

| Year | RUMSONG | PESS SONG | PESSRUM | RUMTIME |
|------|---------|-----------|---------|---------|
| 1955 | 24.7%   | 8.53      | – 2.30  | 1.9%    |
| 1956 | 31.7    | 9.77      | 0.30    | 1.9     |
| 1957 | 28.7    | 8.95      | – 1.06  | 1.9     |
| 1958 | 31.7    | 10.07     | 0.56    | 9.7     |
| 1959 | 27.2    | 8.57      | – 1.72  | 9.7     |
| 1960 | 32.0    | 10.29     | 0.81    | 13.2    |
| 1961 | 33.4    | 10.08     | 0.93    | 17.0    |
| 1962 | 25.4    | 8.83      | – 1.89  | 13.5    |
| 1963 | 27.9    | 8.11      | – 1.97  | 7.7     |
| 1964 | 27.4    | 9.87      | – 0.55  | 5.8     |
| 1965 | 23.2    | 9.14      | – 2.10  | 15.1    |
| 1966 | 30.3    | 10.02     | 0.21    | 19.2    |
| 1967 | 32.5    | 10.81     | 1.37    | 19.2    |
| 1968 | 35.3    | 9.83      | 1.13    | 30.8    |
| 1969 | 28.6    | 11.44     | 1.07    | 25.0    |
| 1970 | 27.2    | 9.26      | – 1.12  | 36.5    |
| 1971 | 36.8    | 10.56     | 2.09    | 25.0    |
| 1972 | 34.2    | 9.58      | 0.67    | 19.2    |
| 1973 | 36.9    | 11.24     | 2.70    | 41.5    |
| 1974 | 25.4    | 8.96      | – 1.78  | 48.1    |
| 1975 | 29.0    | 9.41      | – 0.60  | 30.8    |
| 1976 | 17.5    | 8.79      | – 3.64  | 23.1    |
| 1977 | 18.0    | 10.98     | – 1.63  | 25.0    |
| 1978 | 35.1    | 11.37     | 2.43    | 32.7    |
| 1979 | 36.6    | 10.22     | 1.75    | 47.2    |
| 1980 | 28.9    | 12.59     | 2.14    | 42.3    |
| 1981 | 30.6    | 10.43     | 0.63    | 44.2    |
| 1982 | 27.5    | 11.26     | 0.68    | 50.0    |
| 1983 | 25.9    | 8.17      | – 2.36  | 44.2    |
| 1984 | 26.5    | 10.26     | – 0.41  | 28.3    |
| 1985 | 26.9    | 9.78      | – 0.74  | 40.4    |
| 1986 | 29.2    | 12.40     | 2.04    | 34.6    |
| 1987 | 30.3    | 10.97     | 1.03    | 48.1    |
| 1988 | 25.8    | 10.22     | – 0.60  | 30.8    |
| 1989 | 30.7    | 12.08     | 2.09    | 63.5    |

*Note:* Higher rumination percentages, pessimism scores, and PessRum scores indicate greater rumination, pessimism, and pessimistic rumination.

In this section variables are printed in capitals, and numbers are used at the end of the variable to indicate if it is lagged by one or more years. ‘MA’ indicates a two-year moving average.

Table 2 shows the annual pessimism, rumination, and pessimistic rumination scores for songs, and the annual rumination scores for *Time*

magazine. Pessimistic rumination in popular music lyrics correlated highly with subsequent values of rumination in *Time*, the Index of Consumer Sentiment, personal consumption expenditures, and GNP growth. The two-year moving average of pessimistic rumination correlated  $-0.63$  ( $p < 0.0002$ ) with the moving average of the year-to-year change in GNP (GNPCHG) in the subsequent two years. In a 2SLS analysis, changes in personal consumption expenditures (PERSCONS) predicted GNPCHG, and the Index of Consumer Sentiment (ICS) in turn predicted PERSCONS. One more step removed, pessimistic rumination and rumination in *Time* predicted consumer sentiment.

### *Correlational results*

Rumination in *Time*, PESSRUM in songs, and its components of pessimism and rumination correlated strongly with: subsequent economic growth, the occurrence of a recession year, and the hypothesized mediating variables of consumer optimism and personal consumption expenditures. All correlations are reported with significance levels in parentheses.

RUMTIME correlated negatively with concurrent values of: GNPCHG ( $r = -0.35$ ,  $p < 0.05$ ), the ICS ( $r = -0.68$ ,  $p < 0.0001$ ), and PERSCONS ( $r = -0.40$ ,  $p < 0.02$ ). Fig. 1 shows the negative relationship of RUMTIME and concurrent GNPCHG from 1955 to

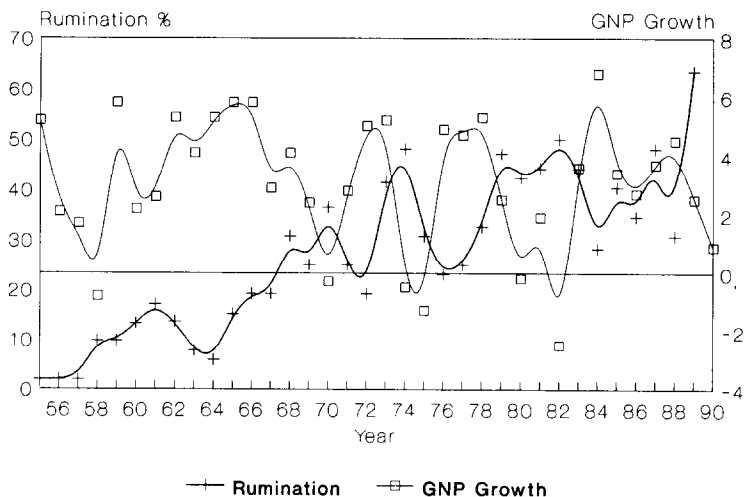


Fig. 1. Rumination on the cover of *Time* magazine correlates negatively with concurrent GNP growth.

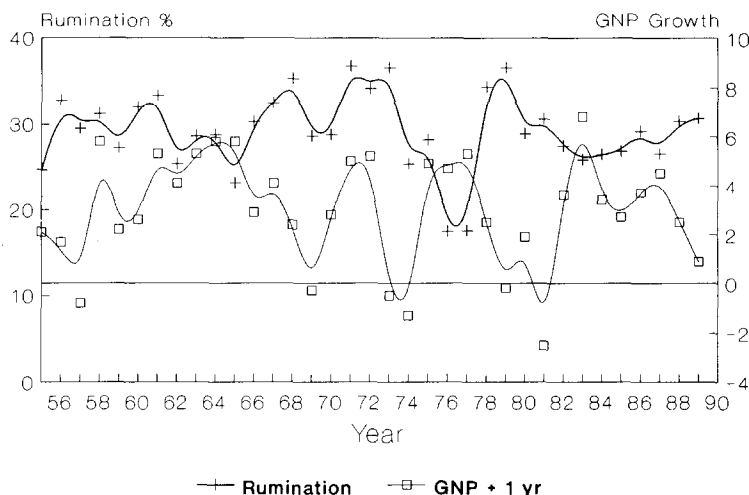


Fig. 2. Rumination in popular songs correlates negatively with GNP growth one year later.

1989. One can see a trend toward increasing rumination in *Time*. The peak rumination, reached in 1989, has yet to manifest itself in a deteriorating GNP performance.

The two-year moving average of rumination in songs (MA RUMSONG) correlated  $-0.38$  ( $p < 0.02$ ) with GNPCHG in the subsequent year,  $-0.45$  ( $p < 0.01$ ) with the moving average of GNPCHG in the subsequent two years, and  $0.50$  ( $p < 0.004$ ) with the occurrence of a RECESSION YEAR two years later (dummy-coded as a 0 for positive real GNP growth and 1 for flat or negative growth). The negative relationship between RUMSONG and the subsequent year's GNPCHG is evident by visual inspection of fig. 2. The rumination moving average also correlated  $-0.36$  ( $p < 0.04$ ) with the Index of Consumer Sentiment for the subsequent year, and the same figure for the moving average of the ICS in the following two years. And it correlated  $-0.41$  ( $p < 0.02$ ) with the change in personal consumption expenditures in the subsequent year.

The two-year moving average of pessimism in songs (MA PESS SONG) lagged one year correlated  $-0.46$  ( $p < 0.009$ ) with GNPCHG,  $-0.58$  ( $p < 0.0007$ ) with the moving average of GNPCHG in the subsequent two years,  $0.75$  ( $p < 0.0001$ ) with RUMTIME,  $-0.57$  ( $p < 0.0007$ ) with the ICS,  $-0.50$  ( $p < 0.005$ ) with PERSCONS, and  $0.30$  ( $p < 0.10$ ) with a RECESSION YEAR. It also correlated  $-0.37$  ( $p <$

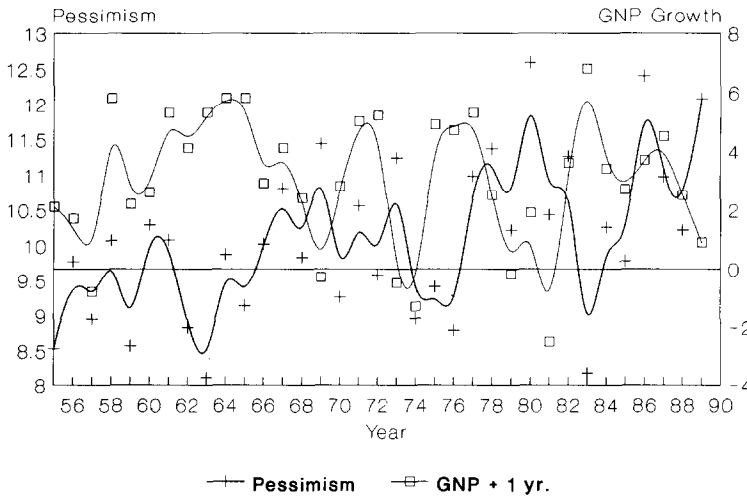


Fig. 3. Pessimism in popular songs correlates negatively with GNP growth one year later.

0.08) with the concurrent two-year moving average of the change in the ICS. Fig. 3 shows the inverse relationship between pessimism and GNP growth in the subsequent year.

The two-year moving average of PessRum (MA PESSRUM) correlated  $-0.63$  ( $p < 0.0002$ ) with the moving average of GNP growth in the subsequent two years,  $0.46$  ( $p < 0.008$ ) with a recession in the following year,  $-0.56$  ( $p < 0.001$ ) with PERSCONS in the following year,  $-0.52$  ( $p < 0.002$ ) with the concurrent moving average of the ICSCHG,  $-0.54$  ( $p < 0.002$ ) with the ICS in the following year, and  $0.54$  ( $p < 0.002$ ) with RUMTIME in the following year. It also correlated  $0.53$  ( $p < 0.005$ ) with the MA GNPCHG five years later, and  $-0.50$  ( $p < 0.006$ ) with the moving average of itself lagged four years, suggesting a negative feedback effect in which PESSRUM predicts lower PESSRUM and higher GNP four to five years later, in contrast to the one- to two-year prediction of slowed growth.

Fig. 4 shows the negative relationship of MA PESSRUM with GNPCHG in the following year. Fig. 5 shows the positive relationship of MA PESSRUM with GNPCHG five years later.

PESSRUM, RUMSONG, PESS SONG, and RUMTIME correlated significantly with the annual average of consumer confidence as measured by the Conference Board's nationwide surveys since 1969, providing an independent validation of the predictive relationship between PESSRUM and consumer confidence.

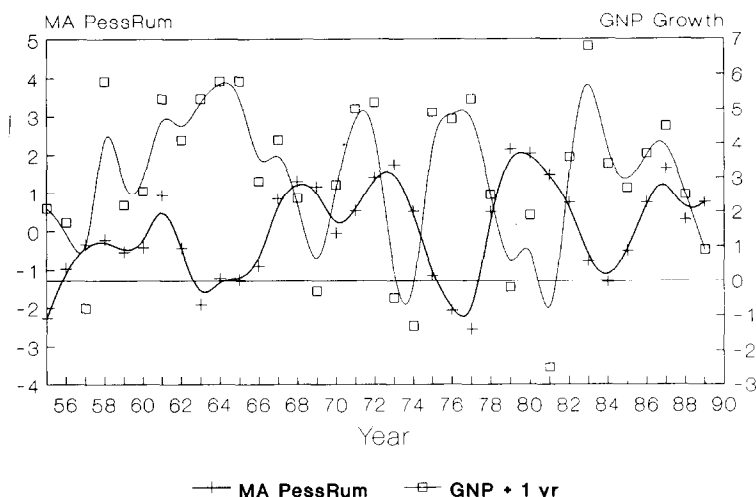


Fig. 4. The two-year moving average of pessimistic rumination in popular songs correlates negatively with GNP growth one year later.

### *Two-stage least squares (2SLS) results*

The two-stage least squares model was specified using a set of structural equations:

$$\text{GNPCHG} = \alpha_1 + \beta_1 \text{PERSCONS} + \epsilon_1.$$

$$\text{PERSCONS} = \alpha_2 + \beta_2 \text{ICS} + \beta_3 \text{ICSCHG1} + \epsilon_2.$$

$$\text{ICS} = \alpha_3 + \beta_4 \text{MA PESSRUM1} + \beta_5 \text{RUMTIME} + \epsilon_3.$$

$$\text{ICSCHG1} = \alpha_4 + \beta_6 \text{MA PESSRUM1} + \epsilon_4.$$

$$\text{MA PESSRUM1} = \alpha_5 + \beta_7 \text{MA PESSRUM5} + \epsilon_5.$$

$$\text{RUMTIME} = \alpha_6 + \beta_8 \text{MA PESSRUM1} + \epsilon_6.$$

This model incorporates several features of the hypothesized model described in the introduction: (1) PESSRUM and RUMTIME as indirect predictors of GNPCHG, via the ICS and change in the ICS, and in turn via PERSCONS; (2) a lagged negative feedback between optimism and economic growth; and (3) PESSRUM in songs having causal precedence over RUMTIME. The PESSRUM composite was used as a main variable rather than RUMSONG and PESS SONG for ease of presentation, and because the summed effect of PESSRUM

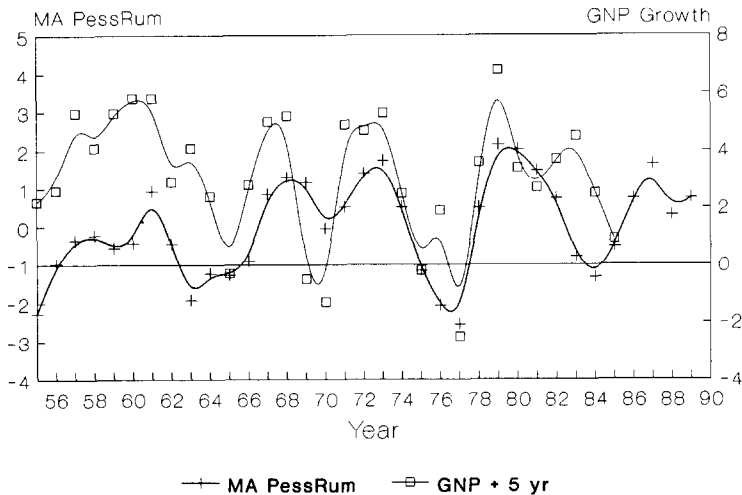


Fig. 5. The two-year moving average of pessimistic rumination correlates positively with GNP growth five years later.

should be superior as a single-term predictor variable to either PESS SONG or RUMSONG.<sup>1</sup>

Estimation was achieved using the 2SLS procedure of SPSS/Trends PC<sup>+</sup>. The instruments were estimated with lagged values of the endogenous variables (specifically, the moving average of PESSRUM, GNPCHG, ICS, ICSCHG, and RUMTIME).<sup>2</sup> Results are shown in table 3.

As one can see, PERSCONS was a strongly significant predictor of GNP growth in the concurrent year. Directly predicting PERSCONS were the absolute level of the ICS in the concurrent year, as well as

<sup>1</sup> An analysis, not reported here, was also performed for the pessimism and rumination components of PessRun taken separately. This analysis revealed, in accord with the analysis reported here, that RUMTIME is the main psychological predictor of ICS, while PESS SONG is the key predictor of *changes* in the ICS and of RUMTIME. RUMSONG, and changes in rumination, appear more critically related than PESS SONG not to continuous changes in GNP but rather the binary state of whether or not the nation enters a recession. This topic will be discussed in a later paper.

<sup>2</sup> MA PESSRUM with lags of 3, 4, and 5 years were used, MA GNPCHG1, MA ICS 1, ICSCHG3, and MA RUMTIME with lags of 1 and 3. These variables were selected because of their utility in estimating the endogenous variables, as determined by separate regression analyses.

Table 3  
Equations in two-stage least squares model.

| Independent variable                                                                                                                                            | Parameter estimate | <i>t</i> | Significance |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------|--------------|
| <i>GNP change regressed on change in personal consumption expenditures</i>                                                                                      |                    |          |              |
| PERSCONS                                                                                                                                                        | 1.33               | 6.24     | 0.0001       |
| Constant                                                                                                                                                        | -1.46              |          |              |
| $R^2 = 0.61$                                                                                                                                                    | $F(1, 25) = 39.0$  |          | $p < 0.0001$ |
| <i>Change in personal consumption expenditures regressed on absolute levels of the index of consumer sentiment, and change in the ICS in the preceding year</i> |                    |          |              |
| ICS                                                                                                                                                             | 0.09               | 3.87     | 0.001        |
| ICSHG1                                                                                                                                                          | 0.08               | 2.33     | 0.03         |
| Constant                                                                                                                                                        | -4.35              |          |              |
| $R^2 = 0.62$                                                                                                                                                    | $F(2, 24) = 19.2$  |          | $p < 0.0001$ |
| <i>Index of Consumer Sentiment regressed on the two-year moving average of pessimistic rumination in the preceding year, and rumination in Time</i>             |                    |          |              |
| MA PESSRUM1                                                                                                                                                     | -3.83              | -1.91    | 0.07         |
| RUMTIME                                                                                                                                                         | -0.48              | -2.84    | 0.01         |
| Constant                                                                                                                                                        | 99.97              |          |              |
| $R^2 = 0.50$                                                                                                                                                    | $F(2, 24) = 11.8$  |          | $p < 0.0005$ |

*Change in the ICS regressed on the concurrent moving average of PessRun*

|              |                             |       |       |
|--------------|-----------------------------|-------|-------|
| MA PESSRUM1  | -4.15                       | -2.42 | 0.025 |
| Constant     | 0.25                        |       |       |
| $R^2 = 0.19$ | $F(1, 25) = 5.8, p < 0.025$ |       |       |

*Moving average of PessRun regressed on itself lagged five years*

|              |                            |       |      |
|--------------|----------------------------|-------|------|
| MA PESSRUM5  | -0.51                      | -2.86 | 0.01 |
| Constant     | 0.08                       |       |      |
| $R^2 = 0.25$ | $F(1, 25) = 8.2, p < 0.01$ |       |      |

*Rumination in Time regressed on the moving average of PessRun in the preceding year*

|              |                             |      |       |
|--------------|-----------------------------|------|-------|
| MA PESSRUM1  | 6.16                        | 2.64 | 0.015 |
| Constant     | 29.81                       |      |       |
| $R^2 = 0.22$ | $F(1, 25) = 7.0, p < 0.015$ |      |       |

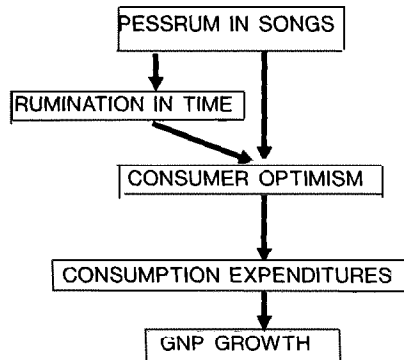


Fig. 6. Path diagram of the predictor variables as indicated by a two-stage least squares model.

*changes* in the ICS one year beforehand.<sup>3</sup> The ICS and ICSCHG1 can in turn be modeled using the two-year moving average of PESSRUM1, which predicted both. PessRum was more significant as a predictor of ICSCHG than ICS – suggesting it serves best as a leading indicator of *changes* in the direction of consumer sentiment. Alternative models in which ICS and ICSCHG are a function of current GNP growth as well as PessRum yielded non-significant *t* values for the parameter for GNPCHG.

RUMTIME was a more significant predictor than MA PESSRUM1 of the absolute level of the ICS, but MA PESSRUM1 directly predicted RUMTIME, suggesting that rumination about current events may mediate between pessimistic fantasies and consumer pessimism.

MA PESSRUM1 can be modeled using the moving average of PESSRUM from four years earlier. The earlier PESSRUM is a negative predictor of PESSRUM four years later.

These pathways are modeled in the flow chart in fig. 6.

The equations in this model were checked for autocorrelated residuals using the AREG procedure of SPSS/PC, which generates a Durbin-Watson statistic. (The AREG procedure does not estimate the equations using 2SLS, but offers a useful approximation for the purpose of estimating autocorrelated residuals.) A Durbin-Watson be-

<sup>3</sup> Ordinarily, an economist might model PERSCONS as a function of income. However, when GNPCHG is introduced as an independent variable in predicting the changes in PERSCONS, its beta weight is non-significant. Only consumer sentiment was significant, so the model reported here includes consumer sentiment and not national income as a predictor of PERSCONS.

tween 1.5 and 2.5 indicates non-autocorrelated residuals, with 2.0 indicating they are perfectly uncorrelated. The Durbin-Watson's for most of the equations were between 1.9 and 2.0. The equation in which PessRum is predicted from previous values of itself was the only equation with autocorrelated residuals (D-W of 1.28).

## Discussion

Pessimistic rumination in popular songs predicts decreased economic growth with a one- to two-year lead. The hypothesis that this relationship is causal was strengthened by the confirmation by simultaneous equations of a plausible path of influence. The path is that pessimistic rumination in the fantasy-based material of song lyrics predicts changes in the degree to which Americans worry about real-world events. PessRum in songs predicts rumination about bad events on the cover of *Time* magazine and consumer pessimism in nationwide surveys. Consumer pessimism in turn predicts decreased personal consumption expenditures, which strongly predicts decreased GNP growth.

What alternative models were not confirmed by the two-stage least squares analysis? Pessimistic rumination was not confirmed as a direct predictor of GNP growth, only as an indirect predictor via consumer sentiment and personal consumption expenditures. This result is reassuring, for the theory being tested specifies a model in which the relationship between pessimistic rumination and GNP growth is mediated by traditional macroeconomic variables.

One model that was not tested might specify pessimistic rumination as a factor in business investment, with an influence on GNP growth via investment as well as consumption. An intriguing area for further research would be to test the relationship of pessimistic rumination with business investment and investment anticipation surveys.

Another possible model is a 'mere correlation' model, in which the absence of a plausible causal pathway would suggest that the pessimistic rumination–GNP relationship is non-causal.

One variation of the 'mere correlation' model states that pessimistic rumination is a proxy for underlying economic conditions. The mere correlation hypothesis cannot be thoroughly ruled out because we have not taken into account all macroeconomic variables for which pessimistic rumination might be a proxy. This is a useful area for future

research. However, pessimistic rumination has been ruled out as a proxy for concurrent GNP growth. Concurrent negative GNP growth might predict an increase in pessimistic rumination. However, controlling for psychological variables, current GNP growth did not predict the ICS, rumination in *Time* magazine, or pessimistic rumination in songs.

In accord with a homeostatic model of the national mood, there is a negative autocorrelation of pessimistic rumination over a five-year lag, while there are no comparable significant autocorrelations for GNP growth. Thus, pessimistic rumination's periodicity appears inherently greater than that of GNP growth, raising the possibility that GNP growth cycles may be partly driven by pessimistic rumination's periodicity. There is a negative correlation between prior economic malaise and pessimistic rumination with a one- to three-year lag, but a positive feedback between pessimistic rumination and subsequent malaise with a one to two-year lag. This lends credence to a Schumpeterian and Mitchellian view of the relationship between optimism and the economy, which holds that in the short run a strong economy may be good for people's outlooks but in the longer term it breeds a contrarian shift due to behavioral errors characteristic of optimists. It may also be true that with continued economic growth, a process of enervation builds up in society that at some point must be dissipated through an economic respite.

One goal the present study did not attempt is to see if pessimistic rumination adds to the predictive power of a traditional econometric model of GNP growth. This should be tested. To some extent, traditional econometric variables may serve as proxies for prior pessimistic rumination. As mediators of the pessimistic rumination–GNP relationship, they may carry the shared variance so that adding pessimistic rumination to the model might not improve its predictive power.

Pessimistic rumination may aid in forecasting economic recessions and growth with a longer lead time than traditional econometric models because it is an early signal for changes in personal consumption expenditures and GNP one and two years later, a longer lead than many economic variables. This would be a useful tool, from two perspectives. One is that warning of a possible downturn in consumer sentiment might encourage private firms to guard against problems such as inventory buildups that would threaten later cutbacks and a recession. A second perspective is the possibility of government action.

Government might find ways to prevent consumer sentiment from swinging in an excessively pessimistic direction. Governments could consider the contrasting impact of policies, depending on whether current sentiment is ruminatively pessimistic or unruminatively optimistic. A tax rise in a year following an increase in pessimistic rumination might inhibit consumer spending, with consumers in a ruminative frame of mind, more than if the rise occurs in a year following an optimistic shift. A similar case might obtain for the risk of aborted growth due to a tight money Federal Reserve policy.

Pessimistic rumination also may moderate the impact of exogenous shocks to the economy such as the 1973 OPEC oil embargo and the seizure of Kuwait by Iraq and its aftermath. As a disposition to develop negative expectations and passivity in response to bad events, pessimistic rumination should make consumers more vulnerable to exogenous shocks. In 1973, a recession had just begun when the oil embargo occurred (Makridakis 1982). Thus, the exogenous shock of the embargo, usually hailed as the cause of the recession, can be regarded as having perhaps deepened and lengthened a trough in the business cycle already in progress. Not only may the embargo have had this effect, but pessimistic rumination may have had a deepening and lengthening effect on the response of the U.S. economy to the embargo. In future research I will test whether pessimistic rumination moderates the impact of objective economic events.

An important issue for further research raised by these results is the origin of changes in pessimistic rumination, and how they are propagated so as to influence consumer sentiment. Are changes in public sentiment dictated by the tastes of elites? If this were true, the media might influence the public's psychological style, and rumination in *Time* magazine might predict changes in songs, rather than vice versa as was the case. Thus, although the media set the agenda of *issues* the public will discuss (Noelle-Neumann 1984), the media may reflect the public's *psychological style*.

Then why do consumer optimism and pessimistic rumination fluctuate? Earlier, I suggested homeostatic and other mechanisms, but this remains an intriguing area for further research. One possibility is that the fantasy world of popular culture may serve a compensatory function for the public. When economic times are bad, people will seek a less worrisome world in movies, music, and other domains. When people are feeling strong and confident in the real world, they may be

more prepared to expose themselves to negative fantasies in the popular culture. These compensatory fantasies may then influence people's real-world evaluations (cf. Parrot and Sabini (1990) on compensatory mood regulation).

Whatever the cause of the fluctuations, high pessimistic rumination in the lyrics of popular songs and rumination on the cover of *Time* magazine have predicted changes in consumer optimism for the last 35 years. Via these changes in consumer optimism, they predicted changes in personal consumption expenditures and Gross National Product. An uptrend in pessimism since 1965 was counterbalanced in the 1980s by low rumination – a 'don't worry, be happy' attitude. It is only at the end of the 1980s, in 1989, that rumination climbed back to high levels concurrently with high pessimism. Iraq's invasion of Kuwait thus came at an inopportune moment in the nation's psyche, since pessimistic rumination in songs in 1989 (especially the second half of 1989) reached one of its highest levels in the last 35 years. Rumination in *Time* magazine in 1989 reached a historic high, in nearly two-thirds of the cover stories, in spite of an auspicious year of economic growth and the demise of Communism in Eastern Europe. This suggests a downturn in consumer optimism in 1990, followed by a recession year in 1991, with the recession beginning as early as later in 1990 – a prediction I first made in May 1990 (Zullo 1990).

If the present model is correct, it suggests that the nation is psychologically vulnerable to negative events during the remainder of 1990 and into 1991. Americans are vulnerable in the sense of being ready to ruminate about negative events and take a pessimistic view of the outcomes, to develop a more gloomy outlook on the economy and curtail our consumption. Thus, we have a prospective opportunity to see whether recent increases in PessRum would translate into lowered consumer optimism and a recession year in 1991. The downturn in consumer optimism that began in July 1990 and accelerated with the onset of the Gulf crisis in early August (personal communication from R. Curtin) is the first confirmation of this possibility.<sup>4</sup>

<sup>4</sup> This article was written mainly before the downturn in consumer sentiment and the invasion of Kuwait in 1990. Events have since shown that an economic recession began in the United States in July of 1990.

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